Omar Haddad

Mechanical Engineer

omar.haddad@example.com | +1 313 555 0109
Detroit, Michigan, United States

PROFESSIONAL SUMMARY

Mechanical engineer with eleven years of experience designing production equipment and electromechanical assemblies. Strong in CAD, tolerance analysis and practical validation planning. Works closely with manufacturing and suppliers to improve manufacturability, reliability and safe maintenance access.

CORE SKILLS

SolidWorks / AutoCAD / GD&T / Finite element analysis / Tolerance analysis / Design for manufacturing / Root cause analysis / Prototyping / Test planning / Technical drawings / Supplier coordination / Lean manufacturing

PROFESSIONAL EXPERIENCE

Mechanical Engineer | Ironwill Motion Systems

Start	End	Location / engagement
04/2022	Present	Detroit, Michigan / Full-time

- Designed modular fixture assemblies for three production lines, including drawings, tolerance stacks and service-access reviews.
- Reduced assembly cycle time by 16% through clamp redesign and improved component orientation.
- Led design verification for a motorized inspection rig, documenting 48 functional and durability test cases.
- Worked with suppliers to consolidate machined parts, reducing unit cost by 12% while maintaining performance requirements.

Design Engineer | Ridgehaven Industrial Products

Start	End	Location / engagement
07/2018	03/2022	Detroit, Michigan / Full-time

- Created detailed CAD models and production drawings for sheet-metal and machined assemblies.
- Investigated field failures using dimensional inspection and root-cause workshops.
- Introduced drawing-release checklists that reduced manufacturing clarification requests by 29%.
- Supported prototype builds and documented design changes through controlled engineering releases.

Graduate Mechanical Engineer | Forgewater Equipment Studio

Start	End	Location / engagement
08/2015	06/2018	Grand Rapids, Michigan / Full-time

- Prepared part drawings, test fixtures and bills of materials under senior engineering supervision.
- Conducted vibration and load tests and summarized findings for design reviews.
- Assisted production teams with fit-up issues and corrective actions.

Omar Haddad

Mechanical Engineer | Qualifications and selected work

EDUCATION

Bachelor of Science in Mechanical Engineering

Fictional Greatlake Technical University

09/2011 - 05/2015 | Grade: 3.7 / 4.0

PROJECTS

Modular Inspection Fixture

Design lead | 01/2023 - 10/2023

Designed a quick-change fixture for six product variants and validated repeatability within 0.08 mm; cut changeover time from 27 to 11 minutes.

Methods / tools: SolidWorks, GD&T, Test planning

https://projects.example.com/omar-haddad/project-1

Cooling Enclosure Redesign

Engineer | 04/2020 - 01/2021

Revised airflow paths and mounting features using simulation and bench tests, lowering peak operating temperature by 9 degrees Celsius.

Methods / tools: Finite element analysis, Prototyping

https://projects.example.com/omar-haddad/project-2

CERTIFICATIONS

Credential / issuer	Issued / expiry	Credential ID
Mechanical CAD Professional; Fictional Design Engineering Guild	12/2022 / No expiry	TEST-09-01
Practical Geometric Tolerancing; Fictional Manufacturing Standards Academy	03/2024 / No expiry	TEST-09-02

LANGUAGES

English (Fluent) / Arabic (Native)

PROFESSIONAL CONTRIBUTIONS

- Member of a fictional local engineering mentoring network since 2021.
- Present design-review workshops on tolerance budgeting and supplier communication.

PROFILE LINKS

https://profiles.example.com/omar-haddad

https://portfolio.example.com/omar-haddad/selected-work